

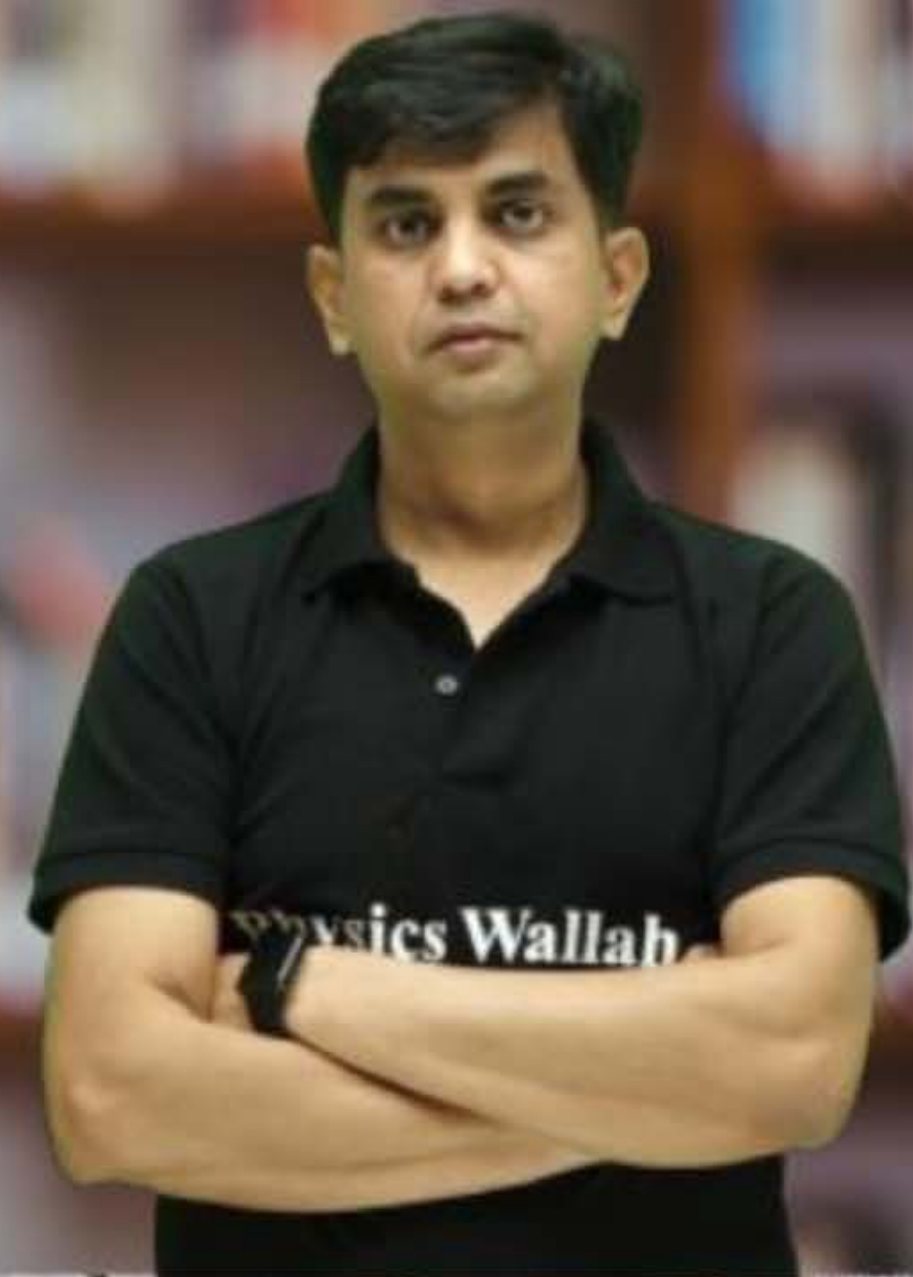
CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 06



By - Abhishek Sir



Recap of Previous Lecture



Topic

CRC

Topic

2D Parity



Topics to be Covered



Topic

2D Parity

Topic

Hamming Code

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
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Topic : 2D Parity



Suppose **Block Size** = **4 x 6** and “**Even Parity**”

Data = “1 0 0 1 1 0 1 1 0 1 1 0 0 1 0”

CASE I : No any error

Receiver

1	0	0	1	1	1	✓
0	1	1	0	1	1	✓
1	0	0	1	0	0	✓
0	1	1	0	0	0	✓
✓	✓	✓	✓	✓	✓	

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 0 1 1 0 0 0 0”

Received Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 0 0 1 1 0 0 0 0”

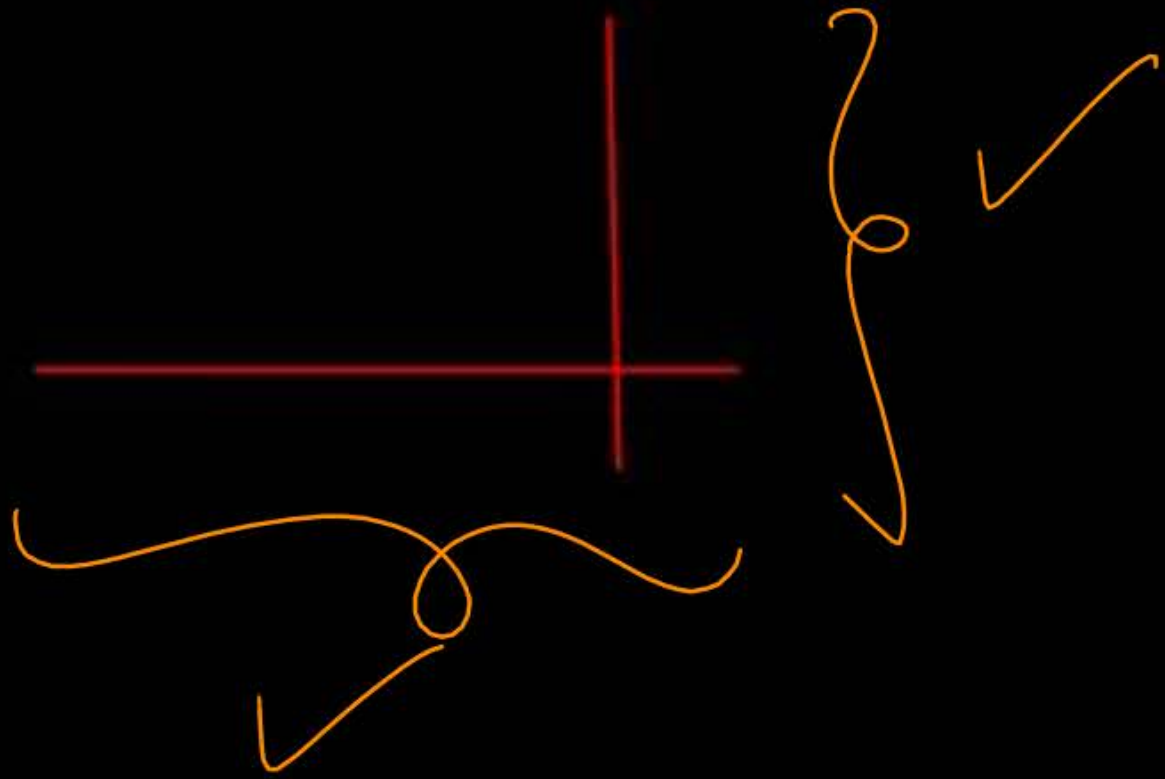
Receiver Concluded : No any Error Detected.



Topic : 2D Parity



CASE I : No any error





Topic : 2D Parity



Suppose **Block Size = 4 x 6** and **“Even Parity”**
Data = “1 0 0 1 1 0 1 1 0 1 1 0 0 1 0”

CASE II : One-bit error

Receiver

1	0	0	1	1	1	✓
0	1	0	0	1	1	←
1	0	0	1	0	0	✓
0	1	1	0	0	0	✓
✓	✓	↑	✓	✓	✓	

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 0 1 1 0 0 0 0”

Received Data = “1 0 0 1 1 1 0 1 0 0 1 1 1 0 0 1 0 0 0 0 0 1 1 0 0 0 0”

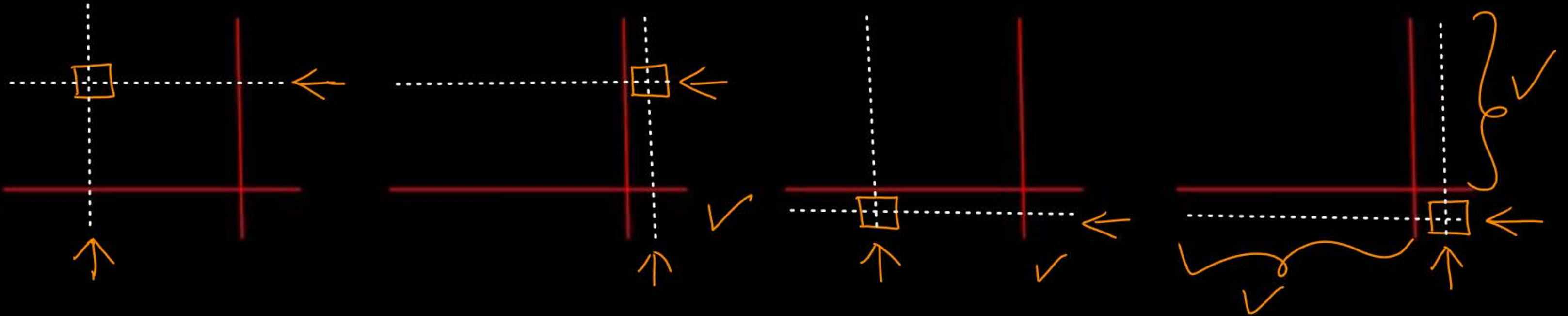
Receiver Concluded : one-bit error detected and corrected successfully



Topic : 2D Parity



CASE II : One-bit error





Topic : 2D Parity



Suppose **Block Size = 4 x 6** and **“Even Parity”**
Data = “1 0 0 1 1 0 1 1 0 1 1 0 0 1 0”

CASE III : Two-bit error

Receiver

1	0	0	1	1	1	✓
0	1	0	0	1	1	←
1	1	0	1	0	0	←
0	1	1	0	0	0	✓
✓	↑	↑	✓	✓	✓	

Sender

1	0	0	1	1	1
0	1	1	0	1	1
1	0	0	1	0	0
0	1	1	0	0	0

Transmitted Data = “1 0 0 1 1 1 0 1 1 0 1 1 1 0 0 1 0 0 0 0 1 1 0 0 0 0”

Received Data = “1 0 0 1 1 1 0 1 0 0 1 1 1 1 0 1 0 0 0 0 1 1 0 0 0 0”

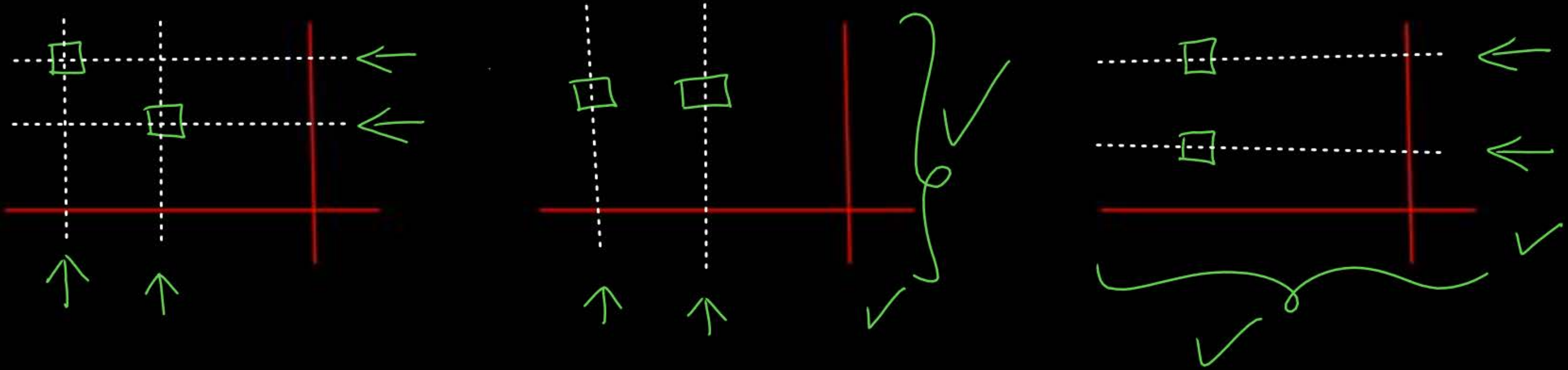
Receiver Concluded : Two-bit Error detected, but unable to correct



Topic : 2D Parity



CASE III : Two-bit error

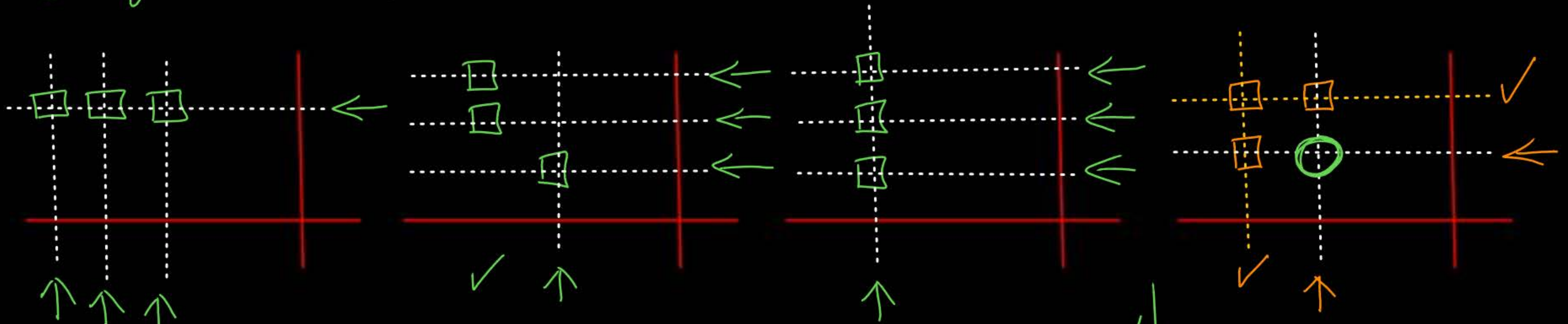




Topic : 2D Parity



CASE IV : Three-bit error



$\Rightarrow \text{Min}(1, 3) = \text{min}^m 3 \text{ bit error}$

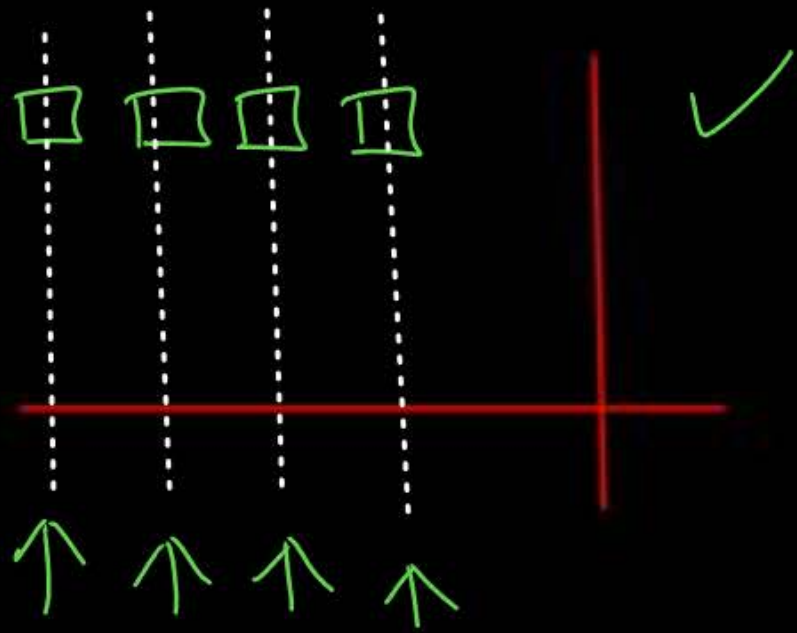
$\Rightarrow$ one bit error detected



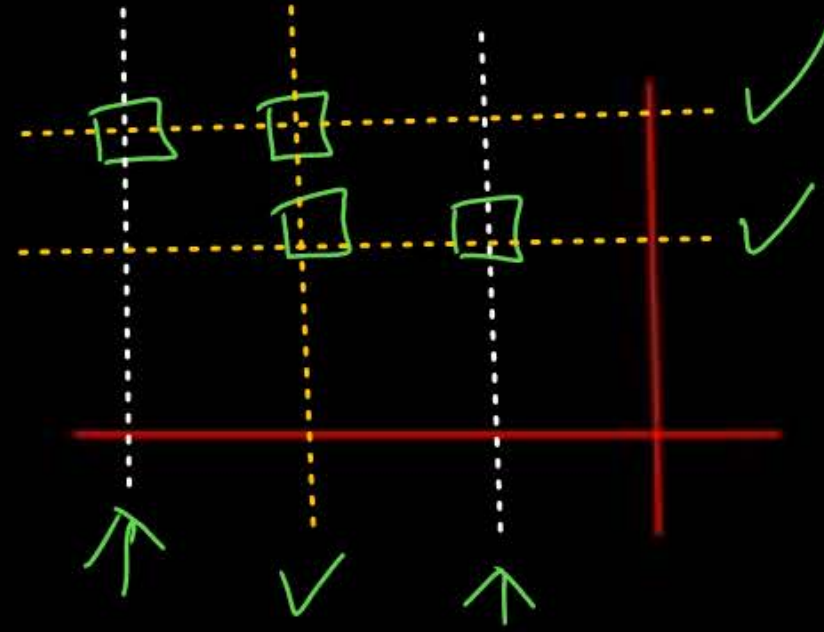
Topic : 2D Parity



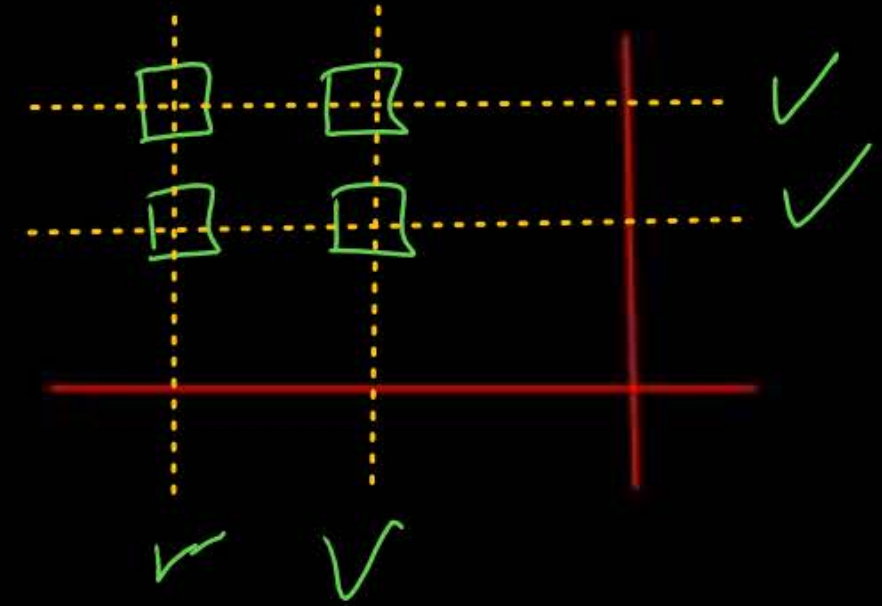
CASE V : Four-bit error



min 4 bit error



min^m 2 bit error



No any Error detected.



Topic : 2D Parity

AT Receiver :-



Minimum number of bits corrupted in the Block

= Maximum [total number of row wise parity unbalanced,
total number of column wise parity unbalanced]

#Q. Data transmitted on a link uses the following 2D parity scheme for error detection:

Each sequence of 28 bits is arranged in a 4×7 matrix (rows r_0 through r_3 , and columns d_7 through d_1) and is padded with a column d_0 and row r_4 of parity bits computed using the Even parity scheme. Each bit of column d_0 (respectively, row r_4) gives the parity of the corresponding row (respectively, column). These 40 bits are transmitted over the data link.

The table shows data received by a receiver and has n corrupted bits. What is the minimum possible value of n ?

[GATE 2008]

(A) 1

(B) 2

(C) 3

(D) 4

ISC
(H.W.)

	d₇	d₆	d₅	d₄	d₃	d₂	d₁	d₀
r ₀	0	1	0	1	0	0	1	1
r ₁	1	1	0	0	1	1	1	0
r ₂	0	0	0	1	0	1	0	0
r ₃	0	1	1	0	1	0	1	0
r ₄	1	1	0	0	0	1	1	0



Topic : 2D Parity



→ Receiver detect and correct “all single bit error” ✓

→ In case of **burst error**,
receiver may able to detect “burst error”



Topic : Hamming Code



- Single bit error-correcting code
- Both transmitter and receiver must agree on same parity
[either “Even Parity” or “Odd Parity”]





Topic : Hamming Code



→ Number of data bits = m

→ Number of parity bits = r

→ Code length (n) = [m + r] bits

→ Hamming (n, m)

Code length
No. of data bits

No of parity bits = (n - m)



Topic : Hamming Code



→ Parity bit placed at position $[R_i] = 2^i$
[where $i = 0, 1, 2, 3 \dots$]

→ Data bits always placed at position $\neq 2^i$

d_7	d_6	d_5	d_4	R_3	d_3	d_2	d_1	R_2	d_0	R_1	R_0
12	11	10	9	8	7	6	5	4	3	2	1
				2^3				2^2		2^1	2^0

R_0	R_1	d	R_2	d	d	d	R_3	d	d	d	d
1	2	3	4	5	6	7	8	9	10	11	12



Topic : Hamming Code



→ Minimum number of parity bits required

$$2^r > (m + r)$$

$$2^r \geq (m + r + 1)$$

→ Minimum code length = 3
[contains only one data bit and two parity bit]

→ Code length $\neq 2^i$
[where $i = 0, 1, 2, 3 \dots$]

$$m + r \neq 2^i$$

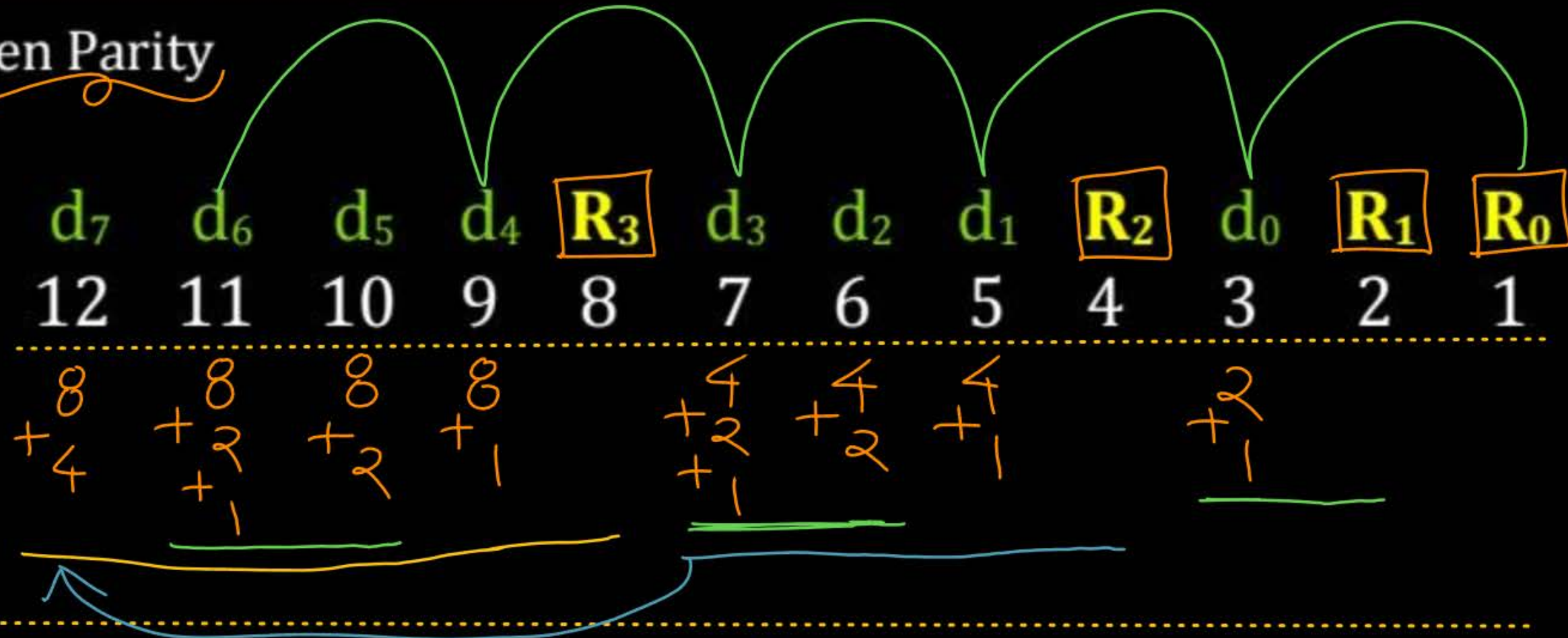
<u>m</u>	<u>r</u>
1	2
2	3
3	3
4	3
5	4
6	4
7	4
11	4
12	5
26	5
27	6



Topic : Hamming Code



→ Even Parity



$$R_0 = d_0 \oplus d_1 \oplus d_3 \oplus d_4 \oplus d_6$$

$$R_1 = d_0 \oplus d_2 \oplus d_3 \oplus d_5 \oplus d_6$$

$$R_2 = d_1 \oplus d_2 \oplus d_3 \oplus d_7$$

$$R_3 = d_4 \oplus d_5 \oplus d_6 \oplus d_7$$

$$[1 = 3, 5, 7, 9, 11]$$

$$[2 = 3, 6, 7, 10, 11]$$

$$[4 = 5, 6, 7, 12]$$

$$[8 = 9, 10, 11, 12]$$

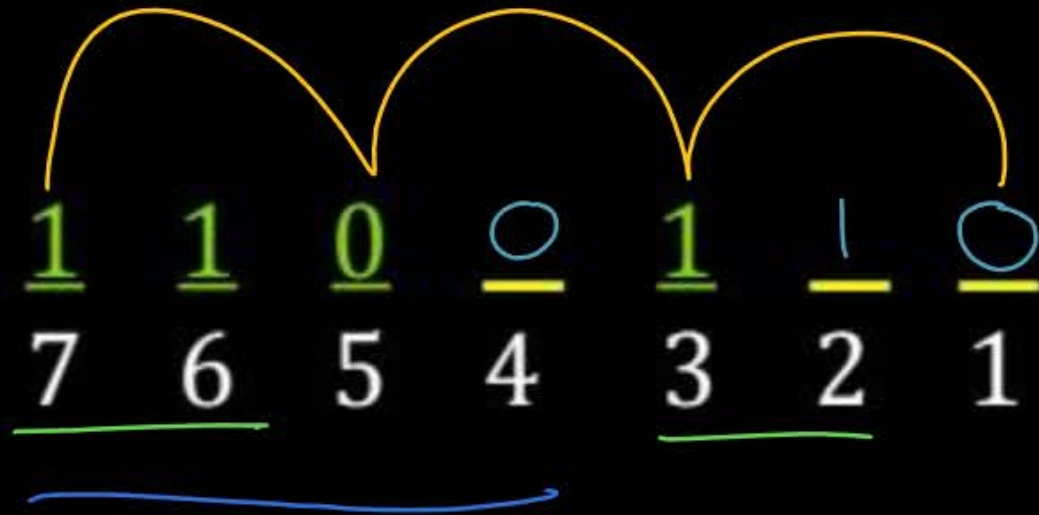


Topic : Hamming Code



Example 1 :

Consider data bits are "1 1 0 1" and "Even Parity", generate hamming code?



$$\textcircled{1} R_0 = 1 \oplus 0 \oplus 1 = 3 \oplus 5 \oplus 7$$

$$\textcircled{2} R_1 = 1 \oplus 1 \oplus 1 = 3 \oplus 6 \oplus 7$$

$$\textcircled{4} R_2 = 1 \oplus 1 \oplus 0 = 5 \oplus 6 \oplus 7$$



Topic : Hamming Code



Example 1 :

Consider data bits are “1 1 0 1” and “Even Parity”, generate hamming code?

<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>
7	6	5	4	3	2	1

Ans : 1100110



Topic : Hamming Code



Example 2 ;

H.W.

Consider data bits are "0 0 1 1" and "Even Parity", generate hamming code?

<u>0</u>	<u>0</u>	<u>1</u>	<u>-</u>	<u>1</u>	<u>-</u>	<u>-</u>
7	6	5	4	3	2	1



2 mins Summary



Topic

2D Parity ✓

Topic

Hamming Code



THANK - YOU

